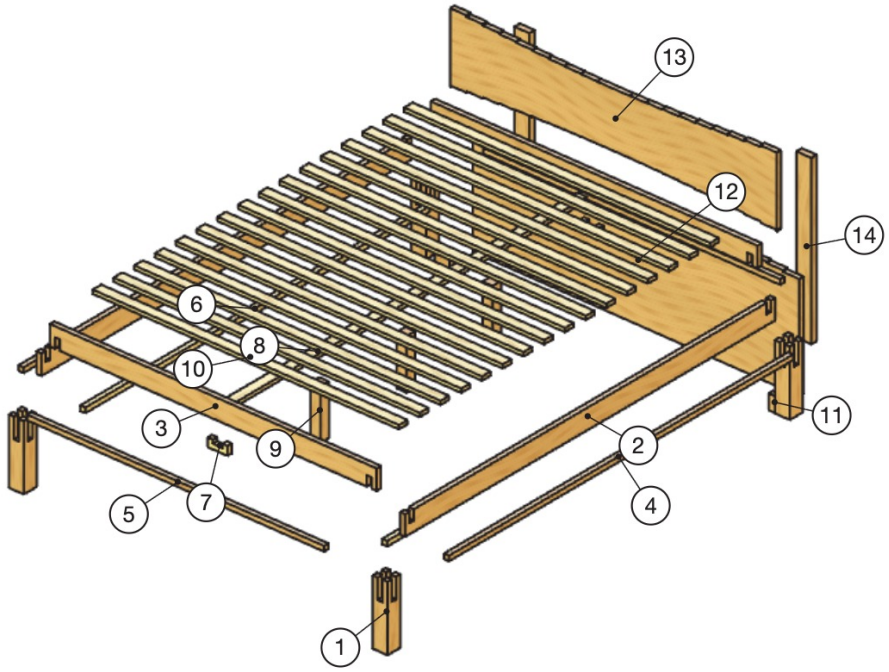
SHOP PLAN

Plan No. MP-01 · Rev A

Castle Leg Bed

Oct 2026

Queen platform frame in quartersawn walnut, split curved headboard



Exploded View

Four 3″ legs carry the frame. Two slots cross in each leg top, the side rail (2) drops in notch-up and the end rail (3) drops over it notch-down, so the laps lock both rails and gravity does the rest. Nothing is bolted or glued at the corners; the frame lifts apart for moving. Walnut cleats (4) (5) fill the step outside each rail, so the frame reads as one flat band flush with the legs all the way around.

WHAT YOU'LL NEED

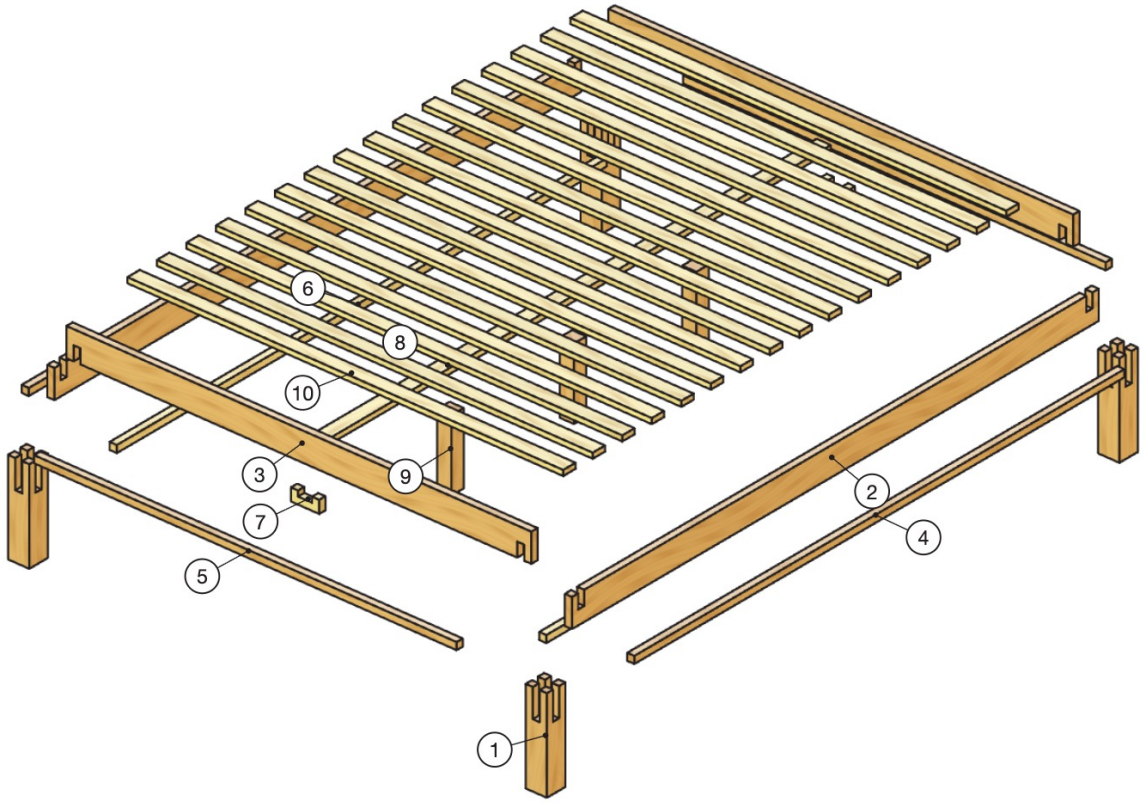
Overall	66" W x 88 1/4" D x 44" H
Wood	Quartersawn walnut & hard maple
Skill	Intermediate
Shop time	4 weekends

- **2 –** 6/4 Walnut, 7-1/4" wide @ 8'
- **2 –** 5/4 Walnut, 7-1/4" wide @ 8'

- **2 –** 10/4 Walnut, 7-1/4" wide @ 8'
- **8 –** 4/4 Hard maple, 7-1/4" wide @ 8'
- **2 –** 5/4 Hard maple, 7-1/4" wide @ 8'
- **14 –** 5/4 Walnut, quartersawn, 7-1/4" wide @ 8'
- **2 –** 8/4 Walnut, rift, 7-1/4" wide @ 8'
- Domino DF 500 with a 5 mm cutter
- A 1/2"-shank pattern or flush-trim bit, 1-1/4" cutting length
- Hardware from the Hard-to-Find Hardware box

Tools: Jointer and planer, or buy stock surfaced four sides; Table saw; Miter saw or crosscut sled; Clamps for the glue-ups; Bandsaw and chisels for the castle slots; Domino joiner; Doweling jig; Drill/driver and countersink; Forstner bit (3/4") for counterbores; Long-series bit (9/32"); Drill press or doweling guide for square holes; Spokeshave; Thin batten for laying out the curve.

The slats sit 1-1/2″ below the rail tops on maple supports and a center rail carried by three walnut posts. The headboard is two quartersawn boards in one plane, split by a 1/4″ gap on a 1-1/2″-rise curve, held by two braces that never reach the floor and bolted through the head rail.

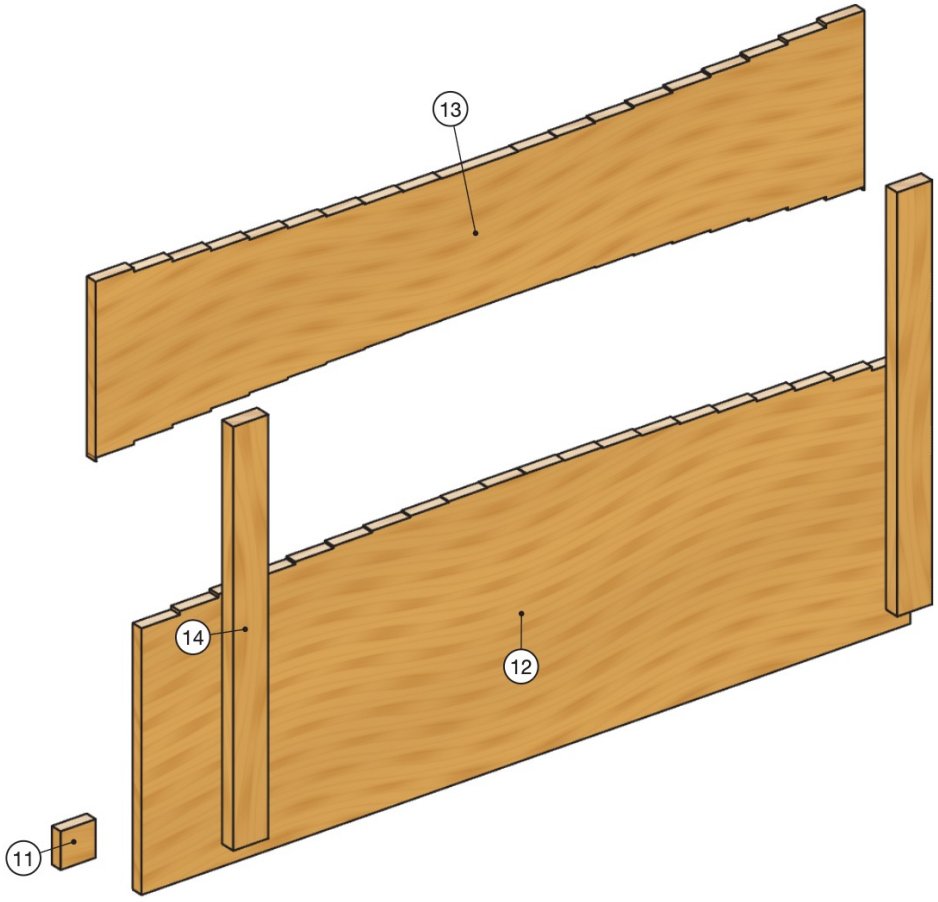


Bed Frame Exploded View

MATERIAL LIST

Bed Frame	T x W x L
1 Legs (4) *Walnut, rift · Two 1-5/8" rift layers, glue line parallel to one slot. 3/4" radius on the outside corner; two inside faces taper to 2-1/4" at the floor (see Leg pattern).*	3" x 3" x 13"
2 Side Rails (2) *Walnut, quartersawn · Lower member of the castle lap: notched from the top.*	1" x 4" x 86"
3 End Rails (2) *Walnut, quartersawn · Upper member of the castle lap: notched from the bottom. Head rail gets six cap-nut seats.*	1" x 4" x 66"
4 Side Cleats (2) *Walnut, quartersawn · Outside the side rails, flush with the rail top and the leg faces.*	1" x 1" x 80"
5 End Cleats (2) *Walnut, quartersawn · Outside the end rails, flush with the rail top and the leg faces.*	1" x 1" x 60"

	T x W x L
6 Slat Supports (2) *Hard maple*	1" x 1" x 80"
7 U Brackets (2) *Hard maple · 2" × 3/4" notch centred in the top edge.*	1" x 1 3/4" x 4"
8 Center Rail (1) *Hard maple · Lies flat in the U brackets, 1/16" short of each end rail.*	3/4" x 2" x 81 7/8"
9 Center Posts (3) *Walnut · A 3/8" dowel glued into each top locates the rail; nothing is screwed into the end grain.*	2" x 2" x 10"
10 Slats (16) *Hard maple · Spaced 2-13/16" apart; one screw into each support.*	3/4" x 2" x 61 7/8"



Headboard Exploded View

MATERIAL LIST

Headboard	T x W x L
11 Spacer Blocks (2) *Walnut · Grain runs side to side, parallel to the rail. Glued to the back of the head rail under the cleat.*	1" x 3" x 3"
12 Headboard Lower Board (1) *Walnut, quartersawn · Three boards edge-glued. Top edge on the template curve, 29-3/4" at the ends rising to 31-1/4".*	1" x 22 1/4" x 66"
13 Headboard Top Board (1) *Walnut, quartersawn · Two boards edge-glued. Top edge dips 1-1/2", bottom edge rises 1-1/2"; 3/4" radius top corners.*	1" x 14" x 66"
14 Headboard Braces (2) *Walnut · Round screw holes next to the gap, slots elsewhere; two bolts at the bottom.*	1 1/4" x 3" x 33"

Legs are listed at finished size before the taper and radius; lay those out from the Leg pattern.
Headboard boards are listed as rectangular blanks; cut the curves from the template.
Saw the castle slots to the actual milled rail thickness.

Castle Leg Bed Hard-to-Find Hardware

Festool Domino 5 × 30 mm beech, 300-pack (494938) (36) ····· **thepowertoolstore.com, Festool dealers**
Cleats to rails.

1/4-20 × 3-1/2" black-oxide connector bolts (Rockler 49153) (4) ····· **rockler.com**
Brace bolts.

1/4-20 × 2-3/4" black-oxide connector bolts (Rockler 47428) (2) ····· **rockler.com**
Center bolts.

1/4-20 cap nuts, black (Rockler 45467) (6) ····· **rockler.com or woodcraft.com**
In a 3/8" seat in the inside face of the head rail, over a washer.

1/4" fender washers, 1" OD, black (2) ····· **Hardware store**
Under the center bolt heads.

3/8" SAE flat washers, 13/16" OD, black (6) ····· **Hardware store**
Under each cap nut, on the inside face of the head rail, to spread the load.

#10 × 2" pan-head wood screws and washers (8) ····· **Hardware store**
Braces to boards.

#8 × 1-1/2" flat-head wood screws (52) ····· **Hardware store**
Supports, brackets, slats.

3/8" walnut or maple dowel, 12" (1) ····· **Woodworking supplier**
Three center-post pins.

Felt or UHMW glides, 1-1/2" (7) ····· **Hardware store**
Legs and center posts.

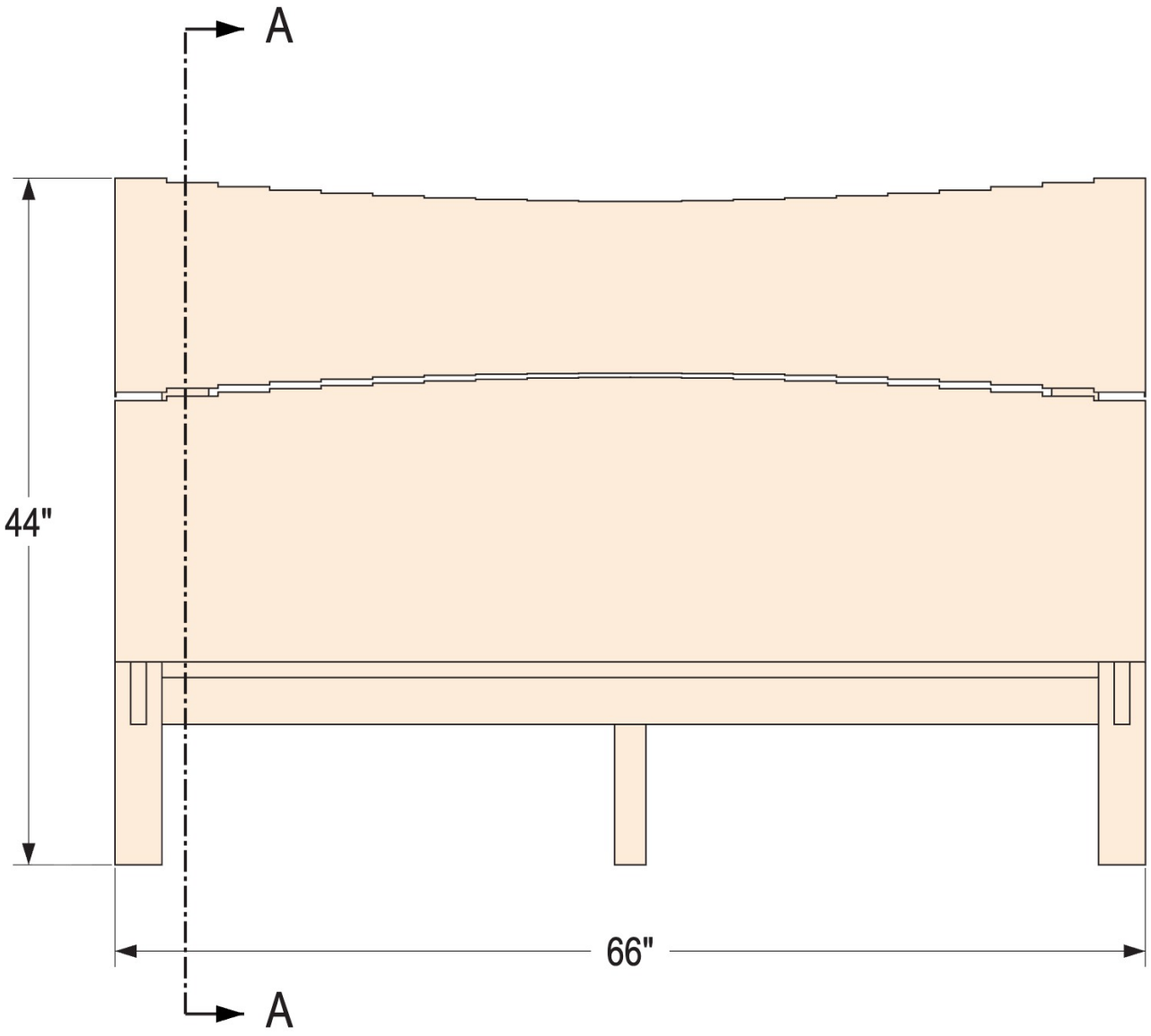
Fasteners, Actual Size

(Lay each fastener on its picture to check it)

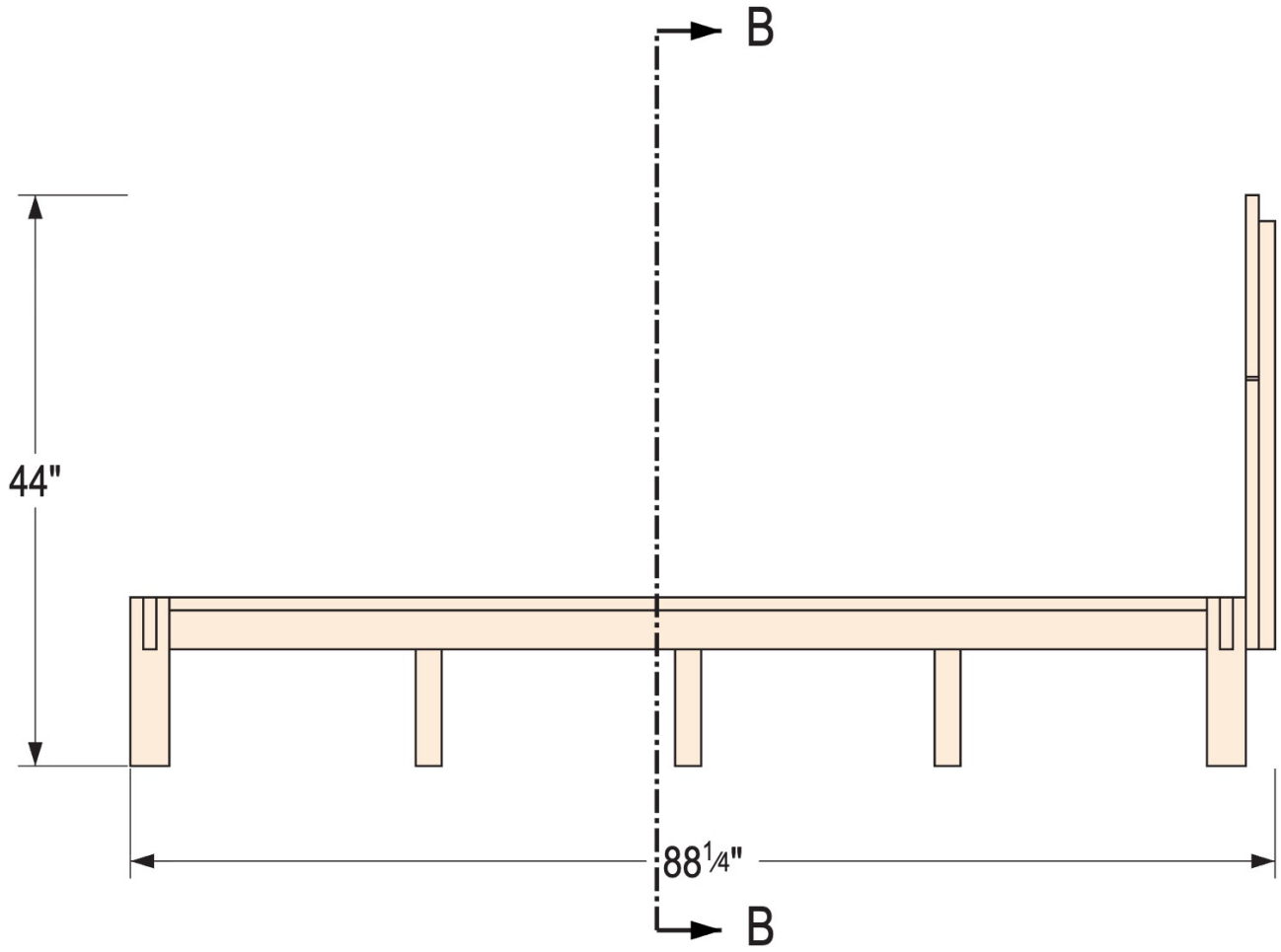


Actual size 7-21/32" x 6-5/32". The PDF prints it at 100% across 2 letter pages with alignment marks; check the 1" square before cutting.

Overall Dimensions
(Front View)

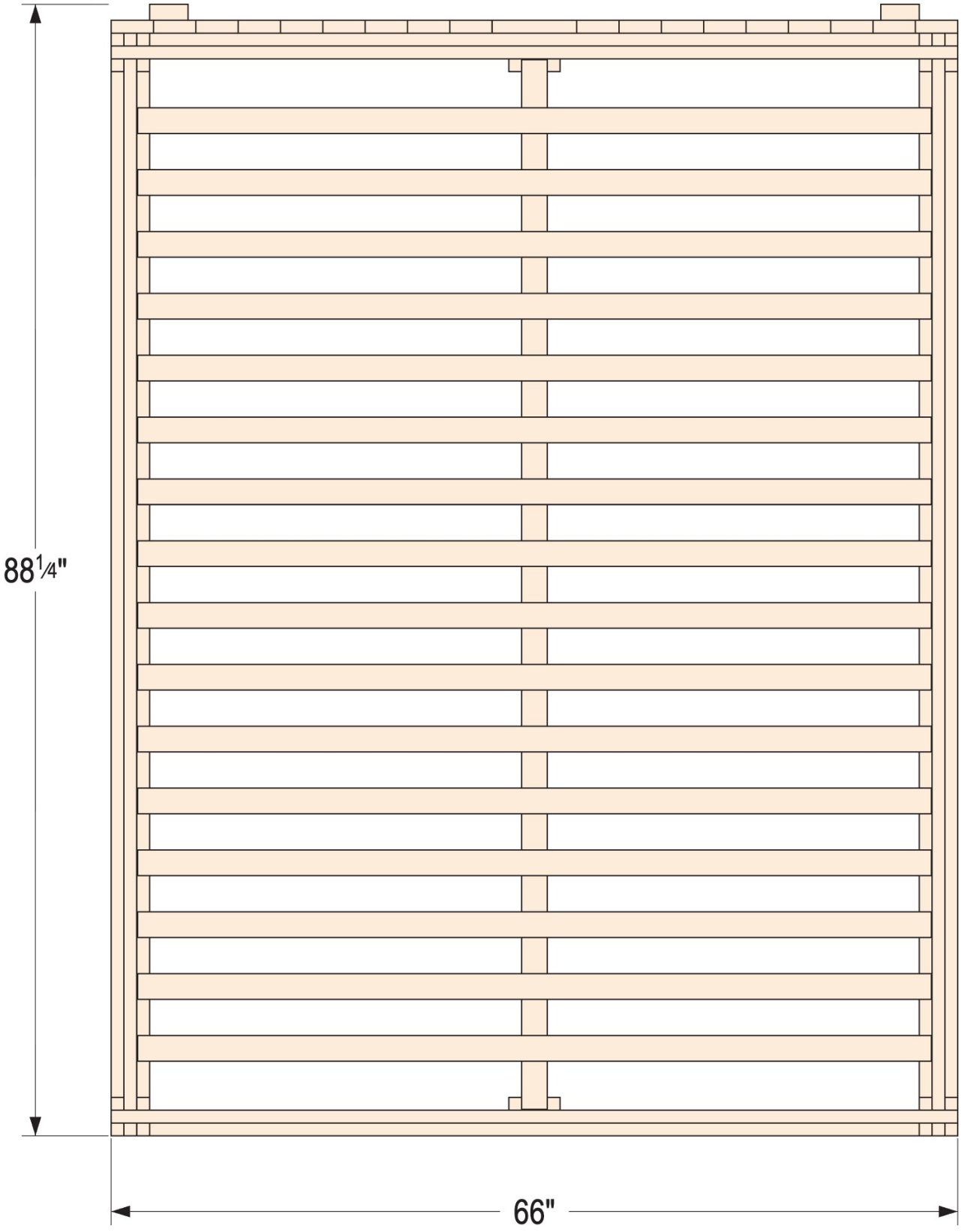


Overall Dimensions
(Side View, from the right)

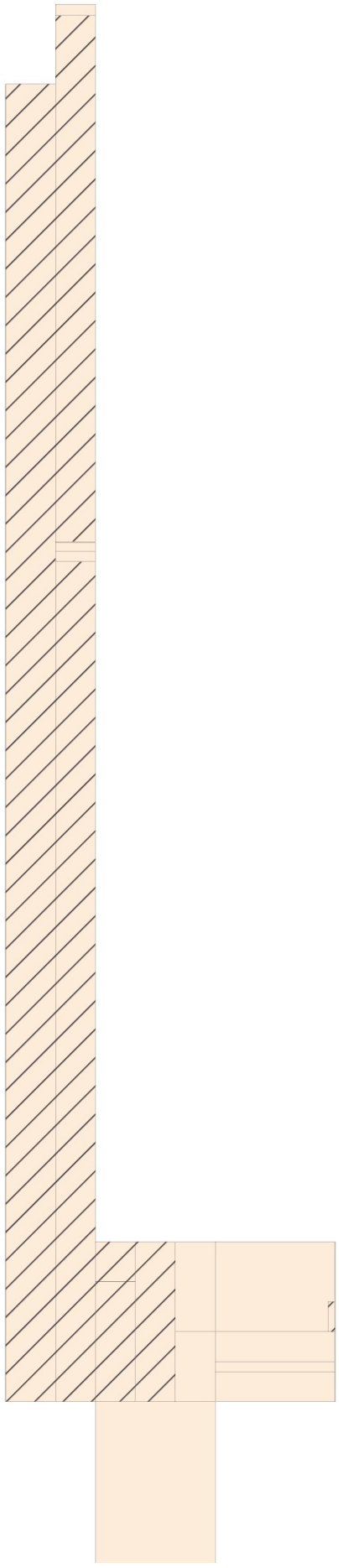


Overall Dimensions

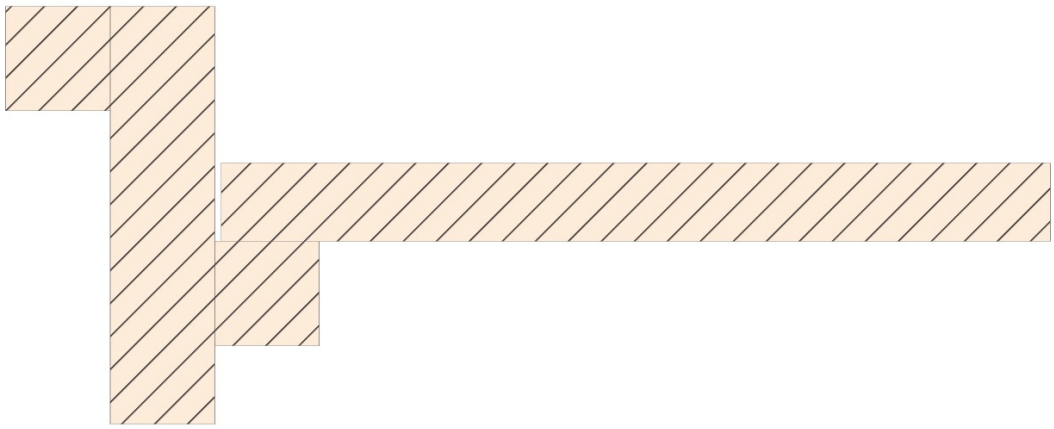
(Top View)



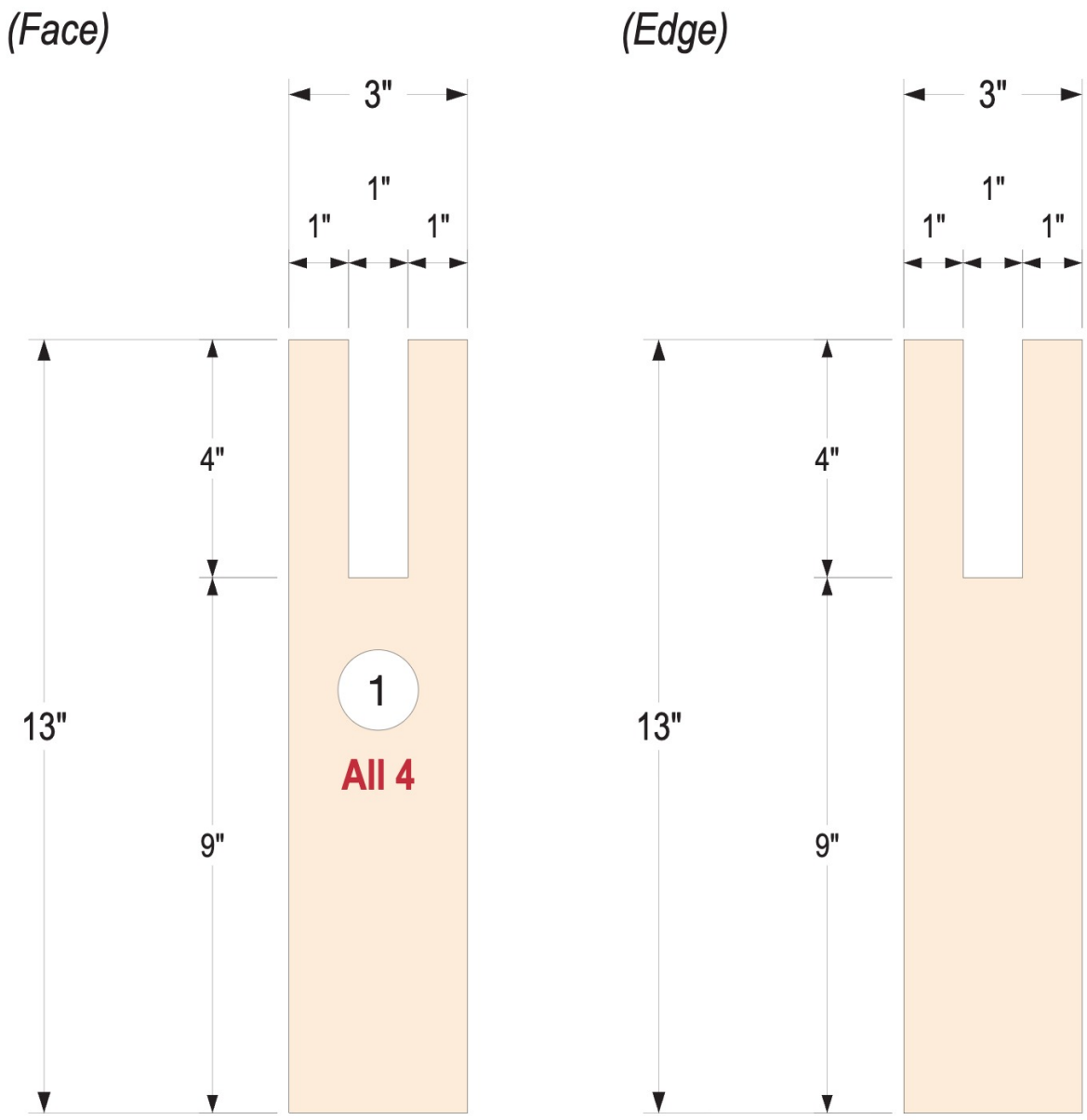
Section A–A
(through the left brace and its bolts)



Section B–B
(through a slat, left side)



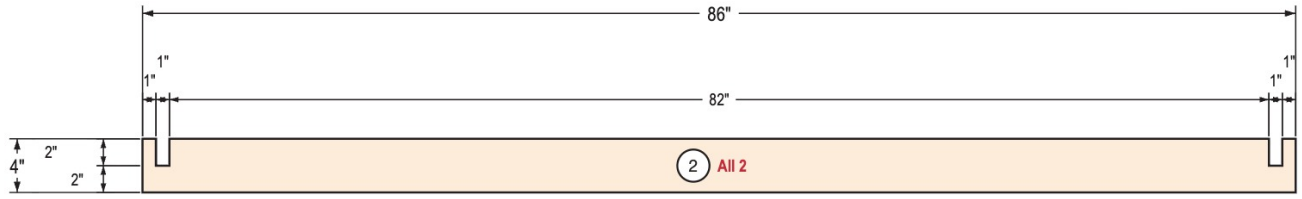
Legs Layout
(Face and Edge Views)



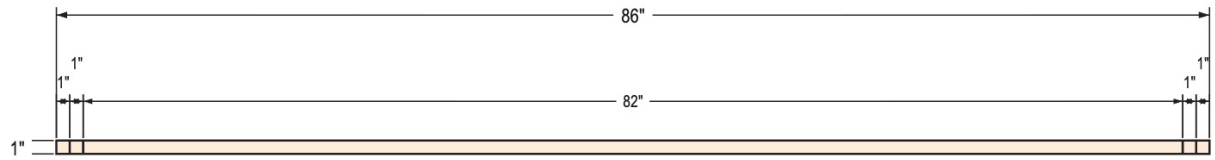
Side Rails Layout

(Face and Edge Views)

(Face)



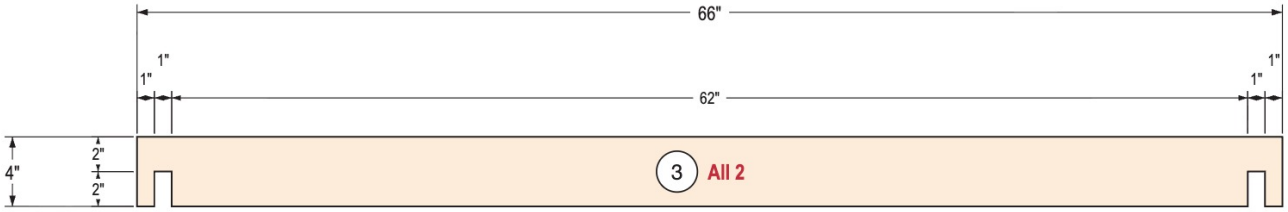
(Edge)



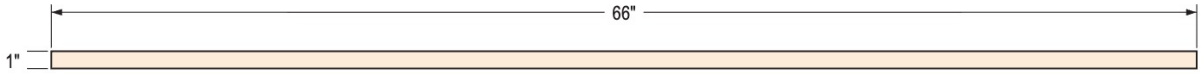
End Rails Layout

(Face and Edge Views)

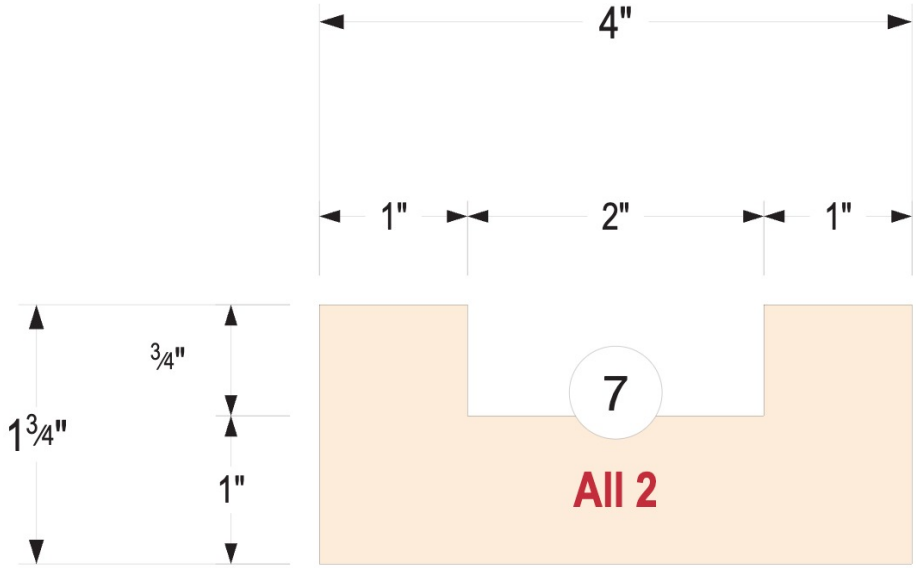
(Face)



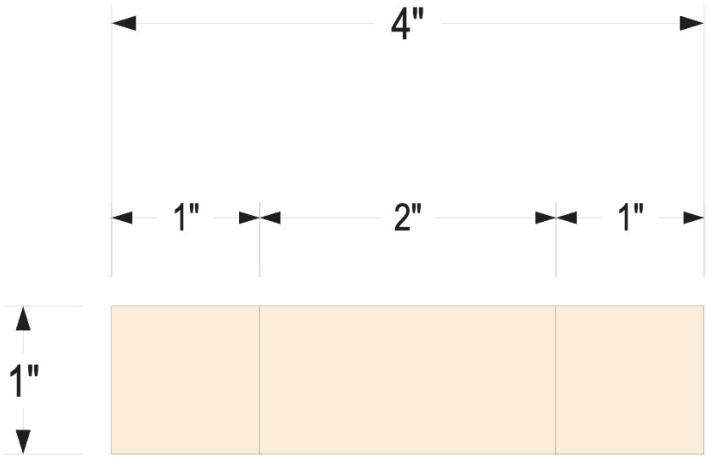
(Edge)



(Face)

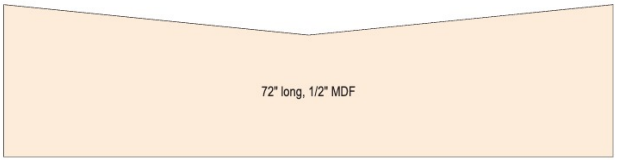


(Edge)



SHOP AIDS

Curve Template
(Router Template)



One 72″ template cuts the top edge of the top board, its bottom edge, and the top edge of the lower board, so the gap stays an even 1/4″.

Part	T x W x L	Material
Template, 1/2" MDF	1/2" x 6" x 72"	MDF
Batten	1/4" x 3/4" x 80"	Straight-grained fir or ash

Using it. Lay out the curve with a thin batten through the offsets on the Full-Size Patterns sheet, fair it with a spokeshave, and flip the template end for end for the second half. Bandsaw 1/8″ outside the line and flush-trim.

For (12) (13)

JOINERY

Every connection in the plan, what holds it, and where it is. **Show** isolates the joint in the exploded view.

No.	Joint	Parts	Held by
J1	**Castle joint, four corners** *Dry, so the frame comes apart. Wax the laps if a corner ever creaks.*	(1) (2) (3)	No glue · None; gravity and the laps
J2	**Side cleats to rails** *Long grain to long grain, parallel.*	(4) (2)	Glue · 5 × 30 mm Dominos about every 8"
J3	**End cleats to rails**	(5) (3)	Glue · 5 × 30 mm Dominos about every 8"; skip any within 1" of a headboard bolt
J4	**Cleat and support ends at the legs** *Cut to fit tight between the legs; not glued to them.*	(4) (5) (6) (1)	No glue
J5	**Slat supports to side rails**	(6) (2)	Glue · #8 × 1-1/2" screws, eight per support
J6	**U brackets to end rails**	(7) (3)	Glue · Two #8 × 1-1/2" screws each
J7	**Center rail in the brackets**	(8) (7)	No glue · None
J8	**Center posts under the rail**	(9) (8)	No glue · 3/8" dowel glued in each post top, loose in the rail
J9	**Slats on supports and center rail**	(10) (6) (8)	No glue · One #8 × 1-1/2" screw each end
J10	**Spacer blocks to head rail** *Grain parallel to the rail.*	(11) (3) (5)	Glue
J11	**Headboard to frame** *Two bolts per brace through brace, lower board, spacer or cleat and rail; two more at the center. Lower-board holes slotted 1/2" tall.*	(12) (14) (11) (5) (3) (2) (1)	No glue · Six 1/4-20 connector bolts into cap nuts
J12	**Braces to boards** *Round holes beside the gap, slots elsewhere, so the 1/4" gap never changes.*	(14) (13) (12)	No glue · #10 × 2" pan-head screws with washers

SHOP CUT LIST

SHOPPING LIST

- **2 –** 6/4 Walnut, 7-1/4" wide @ 8' *(1 in the diagrams, plus spares)*
- **2 –** 5/4 Walnut, 7-1/4" wide @ 8' *(1 in the diagrams, plus spares)*
- **2 –** 10/4 Walnut, 7-1/4" wide @ 8' *(1 in the diagrams, plus spares)*
- **8 –** 4/4 Hard maple, 7-1/4" wide @ 8' *(6 in the diagrams, plus spares)*
- **2 –** 5/4 Hard maple, 7-1/4" wide @ 8' *(1 in the diagrams, plus spares)*
- **14 –** 5/4 Walnut, quartersawn, 7-1/4" wide @ 8' *(11 in the diagrams, counting strips to glue up #12, #13, plus spares)*
- **2 –** 8/4 Walnut, rift, 7-1/4" wide @ 8' *(1 in the diagrams, plus spares)*
- **36 –** Festool Domino 5 × 30 mm beech, 300-pack (494938) *(thepowertoolstore.com, Festool dealers)*
- **4 –** 1/4-20 × 3-1/2" black-oxide connector bolts (Rockler 49153) *(rockler.com)*
- **2 –** 1/4-20 × 2-3/4" black-oxide connector bolts (Rockler 47428) *(rockler.com)*
- **6 –** 1/4-20 cap nuts, black (Rockler 45467) *(rockler.com or woodcraft.com)*
- **2 –** 1/4" fender washers, 1" OD, black *(Hardware store)*
- **6 –** 3/8" SAE flat washers, 13/16" OD, black *(Hardware store)*
- **8 –** #10 × 2" pan-head wood screws and washers *(Hardware store)*
- **52 –** #8 × 1-1/2" flat-head wood screws *(Hardware store)*
- **1 –** 3/8" walnut or maple dowel, 12" *(Woodworking supplier)*
- **7 –** Felt or UHMW glides, 1-1/2" *(Hardware store)*

*Sorted by stock and thickness, then widest to narrowest, then longest to shortest, so parts that share a saw setup sit together. Rough sizes add 1" in length and ¼" in width on solid stock. Tick **R** when the blank is cut and **F** at finished size; ticks are kept in this browser.*

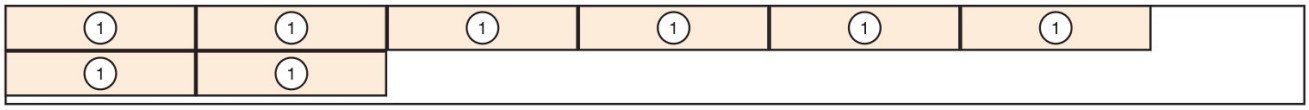
R	F	Part	Qty	Rough T x W x L	Finished T x W x L
10/4 Walnut · mill to 2"					
☐	☐	**9** Center Posts *A 3/8" dowel glued into each top locates the rail; nothing is screwed into the end grain.*	3	2½" x 2¼" x 11"	2" x 2" x 10"
8/4 Walnut, rift · mill to 1½"					
☐	☐	**1** Legs *Glue up from 2 layers · Two 1-5/8" rift layers, glue line parallel to one slot. 3/4" radius on the outside corner; two inside faces taper to 2-1/4" at the floor (see Leg pattern).*	8	2" x 3¼" x 14"	1½" x 3" x 13"
6/4 Walnut · mill to 1¼"					
☐	☐	**14** Headboard Braces *Round screw holes next to the gap, slots elsewhere; two bolts at the bottom.*	2	1½" x 3¼" x 34"	1¼" x 3" x 33"
5/4 Hard maple · mill to 1"					
☐	☐	**7** U Brackets *2" × 3/4" notch centred in the top edge.*	2	1¼" x 2" x 5"	1" x 1¾" x 4"
☐	☐	**6** Slat Supports	2	1¼" x 1¼" x 81"	1" x 1" x 80"
5/4 Walnut · mill to 1"					
☐	☐	**11** Spacer Blocks *Grain runs side to side, parallel to the rail. Glued to the back of the head rail under the cleat.*	2	1¼" x 3¼" x 4"	1" x 3" x 3"
5/4 Walnut, quartersawn · mill to 1"					

☐ ☐	**12** Headboard Lower Board *Three boards edge-glued. Top edge on the template curve, 29-3/4" at the ends rising to 31-1/4".*	1	1 1/4" x 22 1/2" x 67"	1" x 22 1/4" x 66"
☐ ☐	**13** Headboard Top Board *Two boards edge-glued. Top edge dips 1-1/2", bottom edge rises 1-1/2"; 3/4" radius top corners.*	1	1 1/4" x 14 1/4" x 67"	1" x 14" x 66"
☐ ☐	**2** Side Rails *Lower member of the castle lap: notched from the top.*	2	1 1/4" x 4 1/4" x 87"	1" x 4" x 86"
☐ ☐	**3** End Rails *Upper member of the castle lap: notched from the bottom. Head rail gets six cap-nut seats.*	2	1 1/4" x 4 1/4" x 67"	1" x 4" x 66"
☐ ☐	**4** Side Cleats *Outside the side rails, flush with the rail top and the leg faces.*	2	1 1/4" x 1 1/4" x 81"	1" x 1" x 80"
☐ ☐	**5** End Cleats *Outside the end rails, flush with the rail top and the leg faces.*	2	1 1/4" x 1 1/4" x 61"	1" x 1" x 60"
4/4 Hard maple · mill to 3/4"				
☐ ☐	**8** Center Rail *Lies flat in the U brackets, 1/16" short of each end rail.*	1	1" x 2 1/4" x 82 7/8"	3/4" x 2" x 81 7/8"
☐ ☐	**10** Slats *Spaced 2-13/16" apart; one screw into each support.*	16	1" x 2 1/4" x 62 7/8"	3/4" x 2" x 61 7/8"

Rough-cut blanks include 1" extra length and ¼" extra width on solid stock for milling to final size. Grain runs the length of each board. White areas are offcuts.

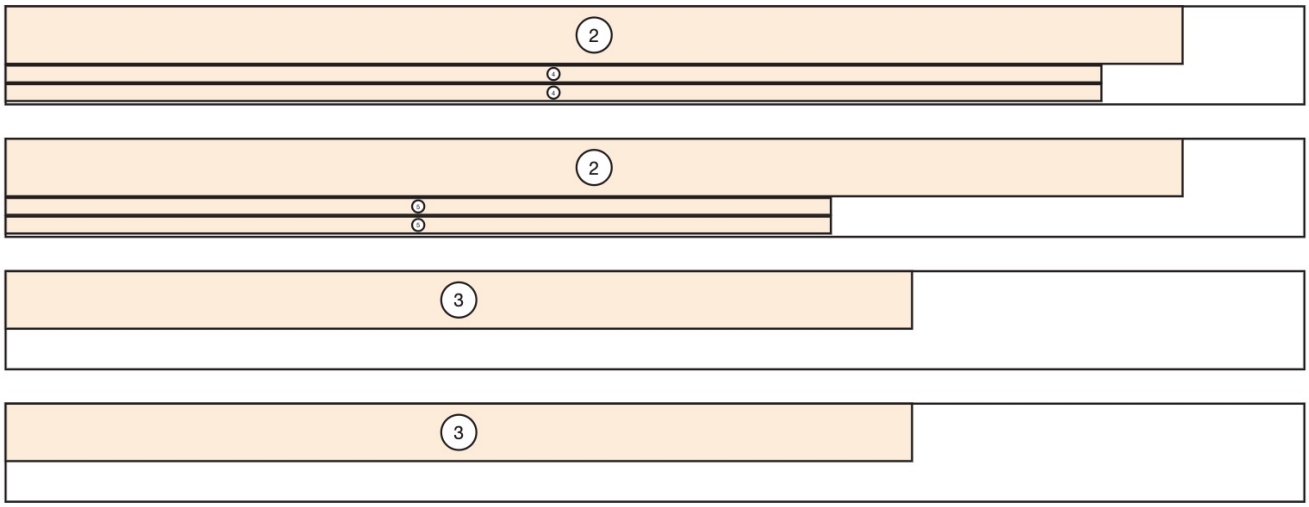
1-1/2" Walnut, rift Cutting Diagram (8/4 stock)

(1 × 7-1/4" × 96" board)



1" Walnut, quartersawn Cutting Diagram (5/4 stock)

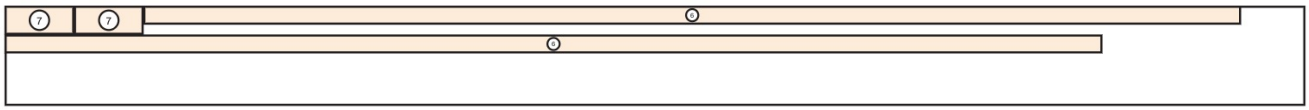
(4 × 7-1/4" × 96" boards)



Too big for this stock, needs a glue-up or wider stock: #12, #13

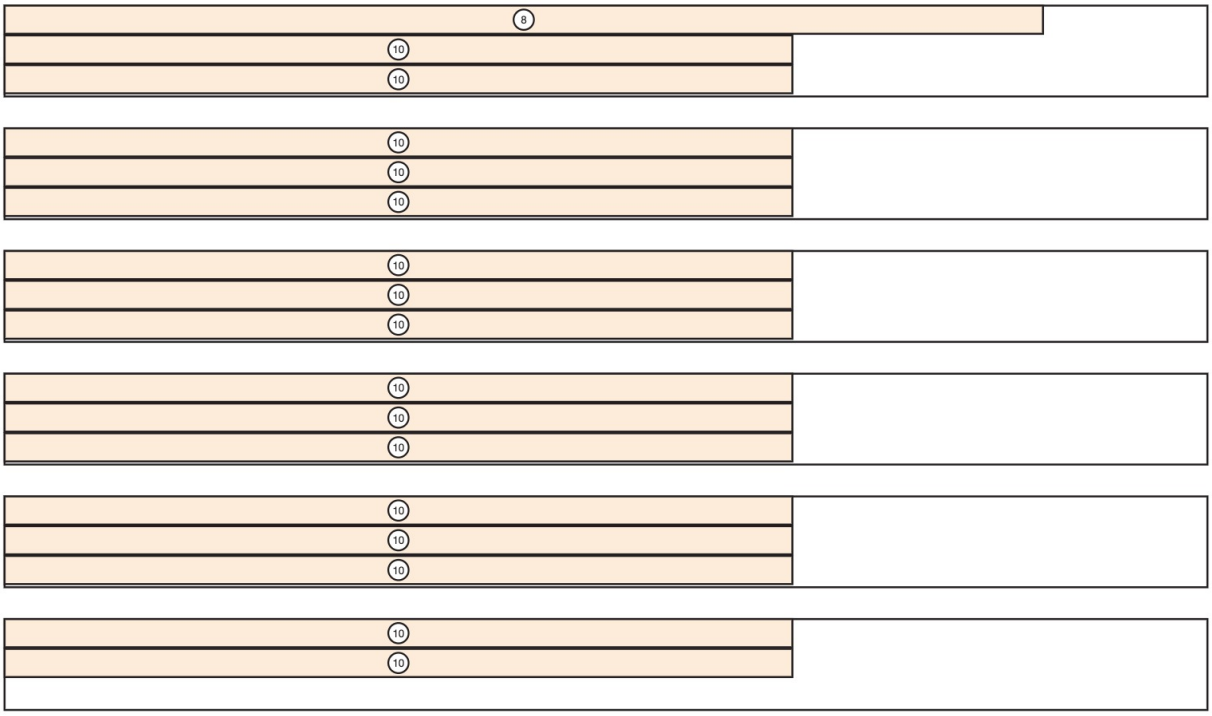
1" Hard maple Cutting Diagram (5/4 stock)

(1 × 7-1/4" × 96" board)



3/4" Hard maple Cutting Diagram (4/4 stock)

(6 × 7-1/4" × 96" boards)



2" Walnut Cutting Diagram (10/4 stock)

(1 × 7-1/4" × 96" board)



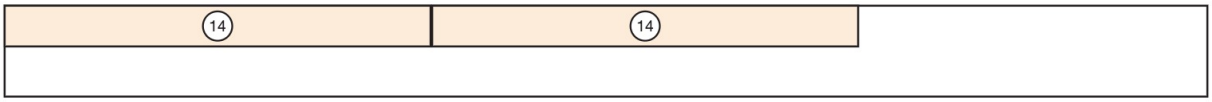
1" Walnut Cutting Diagram (5/4 stock)

(1 × 7-1/4" × 96" board)



1-1/4" Walnut Cutting Diagram (6/4 stock)

(1 × 7-1/4" × 96" board)



Finished	Stock	Material	Board feet (net)	Buy (+30%)	Parts
3/4"	4/4	Hard maple	11.2	14.5	8 10
1"	5/4	Hard maple	1.2	1.6	6 7
1"	5/4	Walnut	0.1	0.2	11
1"	5/4	Walnut, quartersawn	27.0	35.1	2 3 4 5 12 13
1 1/4"	6/4	Walnut	1.7	2.2	14
1 1/2"	8/4	Walnut, rift	3.3	4.2	1
2"	10/4	Walnut	0.8	1.1	9

BUILDING IT

1. Laminate and mill the legs. Glue two 1-5/8″ rift layers for each leg (1), then mill to 3″ square. Rout the 3/4″ radius on the one outside corner while the leg is solid, so the outer tower gets it too.

2. Cut the castle slots. Mill the rails (2) (3) first. Saw both slots 1″ wide and 4″ deep on the bandsaw, clear the waste and pare the floors flat; the rails bear on them.

3. Notch the rails. Side rails (2) notch 2″ from the top, end rails (3) 2″ from the bottom, each 1″ wide and 1″ from the end. Knife the walls from a leg, not a tape.

4. Taper the legs. Taper the two inside faces of each leg (1) from 8″ off the floor to 2-1/4″ at the bottom. Drill the cap-nut and bolt holes later, from the frame.

5. Cleats. Dry-fit the frame. Fit the cleats (4) (5) tight between the legs, cut 5 mm Dominos about every 8″ (skip any within 1″ of 4-1/2″, 22″, 44″ or 61-1/2″ on the head cleat), and glue them flush with the rail tops and the leg faces.

✕ **DON'T** glue the cleats or rails to the legs. Only cleat to rail is glued; the corners must lift apart.

6. Supports and center. Glue and screw the slat supports (6) to the side rails with their bottoms 3/4″ above the rail bottom, and the U brackets (7) to the end rails flush with the rail bottom. Glue a 3/8″ dowel into each center post (9), drill the center rail (8) to drop over them, and set it in the brackets.

7. Slats. Space the slats (10) 2-13/16″ apart and screw each end to its support.

8. Headboard boards. Glue up the lower (12) and top (13) boards. Cut all three curved edges from one template (Shop Aids) with a 1/2″-shank bit, routing downhill from the ends toward the center. Check the 1/4″ gap with spacers before anything else.

9. Braces. With 1/4″ spacers in the gap, screw the braces (14) to both boards: round holes for the screw beside the gap in each board, 1/4″ slots for the others.

✕ **DON'T** use round holes for every brace screw. Each board must grow away from the gap, or the gap closes in summer.

10. Bolt to the frame. Glue the spacer blocks (11) behind the head rail. Clamp the headboard on, drill the six bolt holes, slot the lower board holes 1/2″ tall, and set a 3/8″ washer under each cap nut on the inside face of the rail.

11. Finish and assemble. Finish every part. Stand the legs, drop in the side rails, then the end rails, set the center rail and posts, lay the slats, and bolt on the headboard.

Finishing Schedule *hardwax oil*

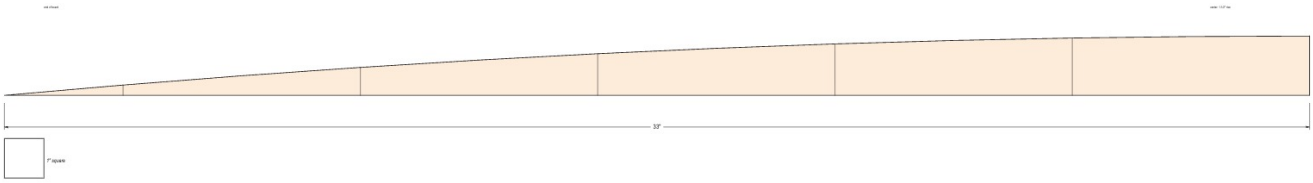
Finish every part before assembly; keep oil off Domino mortises and the spacer glue faces.

No.	Step	Product	How	Dry / recoat
1	**Sand**	150 then 180 grit; end grain to 320	With the grain	none
2	**Raise the grain**	Water	Wipe damp, dry, knock down with 220	1 h
3	**Oil**	Rubio Monocoat Oil Plus 2C, Pure	Thin coat, wipe off all excess within 15 min	24 h, full cure 7 days

*Each drawing here is at actual size. In the PDF it is tiled across letter pages at 100% with a ¼" overlap and alignment marks: tape the pages together, check the 1" square, then trace or glue the pattern to the stock. On screen, **1:1** shows it at your display's nominal size.*

Headboard Curve Template

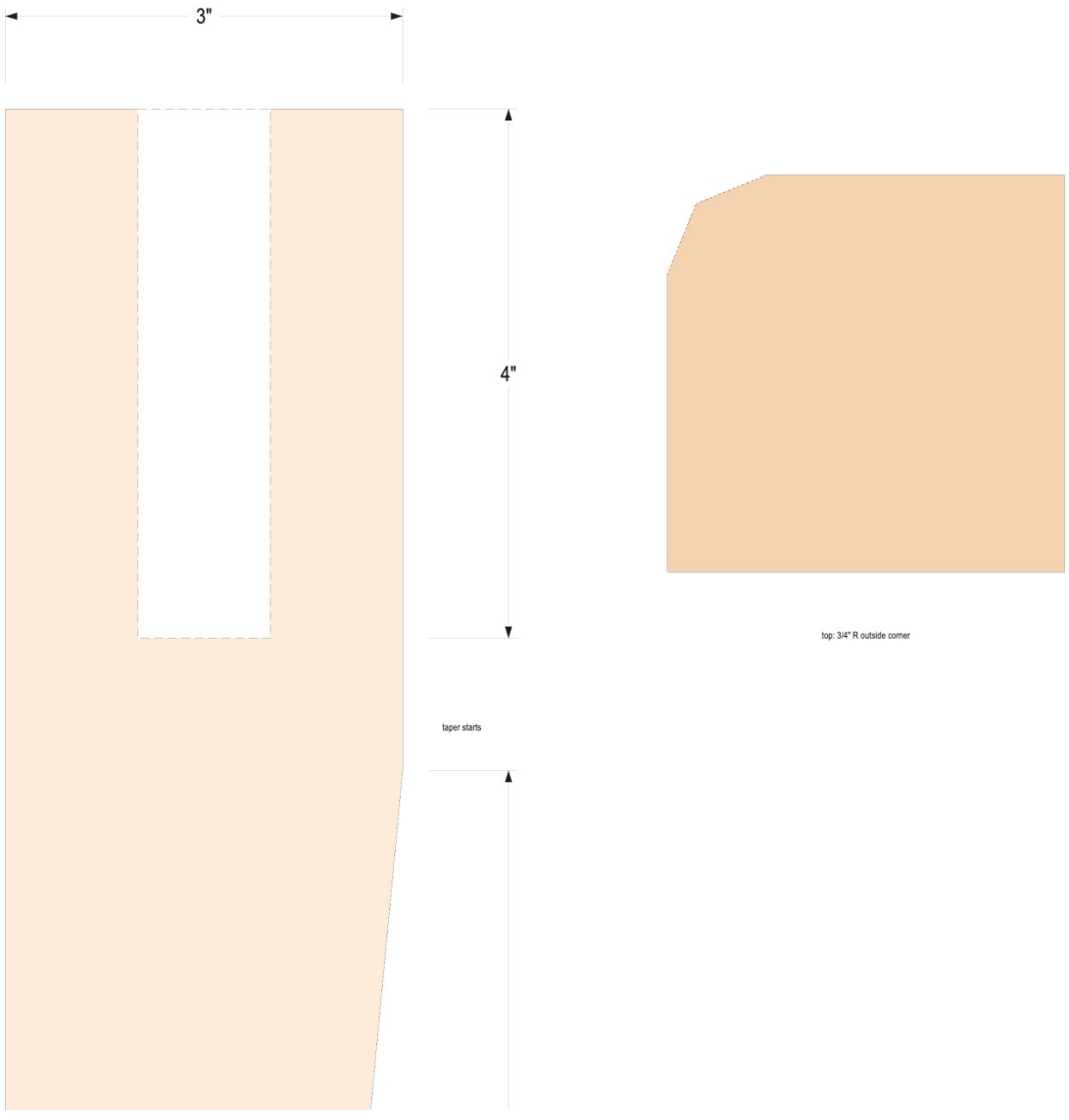
(half template, full size; flip it end for end for the other half)

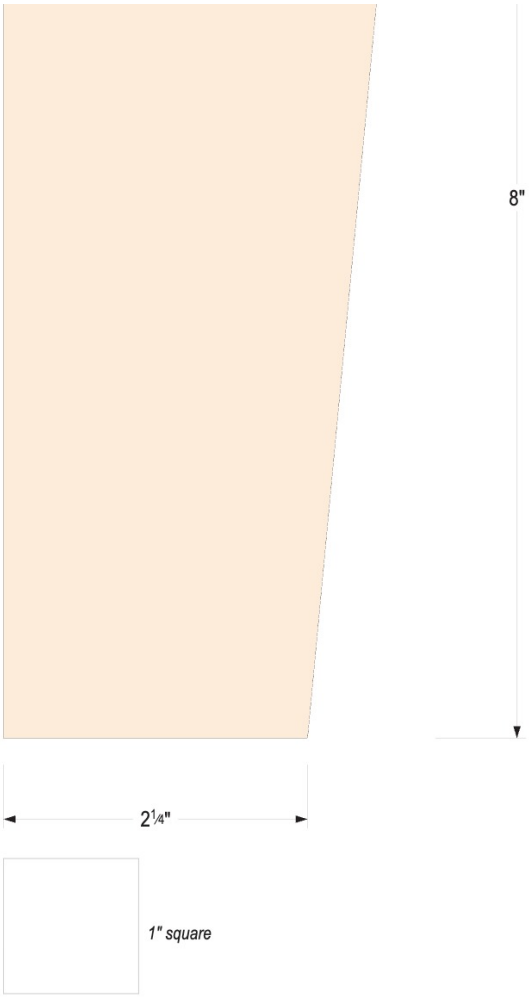


Actual size 33-13/16" x 5-5/32". The PDF prints it at 100% across 5 letter pages with alignment marks; check the 1" square before cutting.

Leg Pattern

(inside face and top view, full size)





Actual size 8-13/16" x 16-1/2". The PDF prints it at 100% across 4 letter pages with alignment marks; check the 1" square before cutting.

Headboard Story Stick

(Full Size, 35" long)



***0"** #3 End Rails, #11 Spacer Blocks, #12 Headboard Lower Board, #14 Headboard Braces · **1-1/4"** brace bolt · **3"** #11 Spacer Blocks · **3-1/2"** bolt · **4"** #3 End Rails · **11"** slot · **19-19/32"** round · **21"** #13 Headboard Top Board · **22-1/4"** #12 Headboard Lower Board · **22-29/32"** round · **32"** slot · **33"** #14 Headboard Braces · **35"** #13 Headboard Top Board*

Actual size 36-5/16" x 4". The PDF prints it at 100% across 5 letter pages with alignment marks; check the 1" square before cutting.

REVISIONS & REVIEW

Every change to this plan gets an adversarial review: each change is walked against the parts it touches and the parts those mate with, and the geometry is re-checked by the model below. A change ships when every line passes or is fixed.

Rev A *Oct 26 2026 · First issue in the woodworking-plans format*

A1 · Whole design, full adversarial review FLAG

Touches ① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩ ⑪ ⑫ ⑬ ⑭ · mates

Check	Finding	Result
Dimensions ① to ⑭ re-derived from PARAMS	Outside 60 + 2 × 3 = 66" × 80 + 6 = 86". Rails 1 × 4 × 86 and 66. Cleats and supports 86 − 6 = 80, end cleats 60. Center rail (84 − 2) − 1/8 = 81-7/8". Posts 10-3/4 − 3/4 = 10". Slats (64 − 2) − 1/8 = 61-7/8", gaps (80 − 32) ÷ 17 = 2-13/16". Lower board 31-1/4 − 9 = 22-1/4" blank, top board 44 − 30 = 14", braces 42 − 9 = 33". All match the cut list and the model.	PASS
J1 castle ① ② ③: proportions, tools, load path	Slots 1" × 4" in a 3" leg leave 1" towers, a third of the leg (rule: a quarter, 3/4" minimum). Laps 2" + 2" = rail height. Slots on the bandsaw (4" deep); rail notches 2" deep, inside a dado stack's 2-1/2". Side rail bears on the slot floor, end rail on the side-rail lap: wood on wood to the floor.	PASS
Leg corner radius on a 1" tower ①	A 3/4" radius removes 0.56 × (1 − π/4) = 0.12 sq in of the 1 sq in outer tower, 12%. The tower is long grain and captured by both rails once assembled. My earlier note called it fragile; the numbers do not support that.	PASS
Side rail size ②	joinery.md asks for 1-1/4" × 6" minimum on a queen side rail; these are 1" × 4". With slats continuous over the center rail, each side rail carries about 3/16 of the load: 112 lb of a 600 lb design load over 80" gives 420 psi bending stress (walnut breaks near 14,600) and 0.08" sag. A 250 lb person sitting mid-rail gives about 0.22" of flex, counting the glued cleat (I 7.2 in^4). Strong enough; it will flex a little when sat on. Going to 1" × 5" would cut that flex by half. Your call.	FLAG
J2 J3 Dominos ④ ⑤ ② ③	5 mm Dominos 15 mm deep into 25 mm parts leave 10 mm walls. Cleat and rail grain run the same way, so a full glue line is safe. Glued cleats sit between the legs, so each rail still drops straight down into its slots.	PASS
J5 J6 supports and brackets ⑥ ⑦	Glued long grain to the rails, with screws as clamps. The model check warns that a 1-1/2" screw is 1.5× the 1" support; a 2" screw would come through the 1" rail, so 1-1/2" (1/2" into the rail) is the longest that fits. The glue line carries the load.	PASS
J8 center posts ⑨ ⑧	The old page screwed the center rail down into the end grain of each post. Now a 3/8" dowel glued 1" into each post top passes up through a hole in the 3/4" rail. The load is the rail bearing on the post end grain; nothing hangs on a screw.	FIXED
J9 slats ⑩ on ⑥ ⑧	Slats end 1/16" from the rail faces, so each bears on 15/16" of the 1" support. Slat screws were #8 × 1-1/4" (1.7× the slat, 1/2" into the support); now #8 × 1-1/2": 2× the slat and 3/4" into the support. The middle screw into the 3/4" center rail is dropped; a 1-1/2" screw would come through it. First slat starts at 5-13/16", clear of the U brackets (2" to 3").	FIXED
J10 spacer blocks ⑪	The old cut sheet gave no grain direction for a 3" × 3" block glued to the rail. Now grain runs side to side, parallel to the rail, so the glue line is long grain to long grain.	FIXED
J11 headboard bolts ⑫ ⑭ ⑪ ⑤ ③: reach and clearance	Brace bolt stack 1-1/4 + 1 + 1 + 1 = 4-1/4"; less the 1/2" counterbore leaves 3-3/4", so the 3-1/2" bolt stops 1/4" short of the rail face, 3/8" (7 threads) into the 5/8" cap nut. Center stack 3", 2-3/4" bolt, same 3/8" engagement. Cap nuts at 4-1/2" sit 1-1/2" from the leg face; the center nuts at 22" and 44" clear the U brackets at 31" to 35". Two bolts per brace, 2-1/4" apart, so the headboard cannot hinge.	PASS

J11 headboard bolts: bearing under the cap nuts	A 100 lb lean at 34" pivots the headboard on the frame at 9": 2,500 in-lb, about 170 lb on each upper bolt. The cap nut flange alone bears on about 0.14 sq in of walnut, 1,200 psi, above walnut's roughly 1,000 psi crushing strength across the grain: it would dent in over time. Added a 3/8" SAE washer (13/16" OD) under each cap nut: 0.40 sq in, 420 psi. The washer top stays 3/32" below the rail top. Pushing the bed against the wall takes most of this load anyway.	FIXED
J12 braces (14) (13) (12): movement	Quartersawn walnut moves about 0.0019 per inch per 1% moisture; over a 4% swing the lower board moves 0.17" and the top board 0.11". Each board is pinned only by the screw beside the gap (31.9" and 28.6"), so the gap stays 1/4". Farthest slotted screw is 9.1" from its pin: 0.07" of travel in a 1/4" slot. The bolts are 18.4" below the lower pin: 0.14" in a 1/2" slot. The model check warns a 2" screw is 1.6× the 1-1/4" brace; the board is only 1" thick, and 2" already leaves just 1/4" of wood in front of the tip.	PASS
Mattress fit	Leg faces are exactly 60" × 80" apart; the rails stand 1" farther out on each side. A standard queen touches only at the four inside leg corners. Measure yours; an oversize mattress needs the legs moved.	PASS
Paperwork	Every bore has a step: B1 to B4 in step 10, B5 in 9, B6 and B8 in 6, B7 in 7. #8 × 1-1/2" count: 16 supports + 4 brackets + 32 slats = 52. Dominos: 10 per side cleat and 8 per end cleat, less any skipped beside the bolts at 4-1/2", 22", 44", 61-1/2": 36 or fewer. Intro: slats 1-1/2" below the rail tops (13 − 11-1/2).	PASS

Model check *recomputed from the drawing data each time this page loads*

CHECK	Bore B5.1 (#10 × 2" pan-head, J12): a 2" screw through 1-1/4" of #14 Headboard Braces is 1.6× the part. Aim for 2× to 3×. (and 7 more copies of this bore) (14)
CHECK	Bore B6.1 (#8 × 1-1/2" flat-head, J5): a 1-1/2" screw through 1" of #6 Slat Supports is 1.5× the part. Aim for 2× to 3×. (and 1 more copy of this bore) (6)
PASS	14 parts: cut list T x W x L matches the modeled geometry.
PASS	No two parts occupy the same space (tenons excepted, inside declared joints).
PASS	12 of 12 declared joints have real contact (or a hardware gap within tolerance).
PASS	All 14 modeled parts connect into one assembly.
PASS	Overall size in the title block matches the model: 66" W x 88-1/4" D x 44" H.

Fasteners, Actual Size · page A1 (1 row x 2)

FULL SIZE: print at 100% ("Actual size"). Overlap 1/4" with the next page and match the marks.



1/4-20 × 3-1/2" connector bolt · 9/32" x 3-1/2" · ×4



1/4-20 × 2-3/4" connector bolt · 9/32" x 2-3/4" · ×2



#10 × 2" pan-head · 3/16" x 2" · ×8



#8 × 1-1/2" flat-head · 5/32" x 1-1/2" ·



3/8" dowel · 3/8" x 1-3/4" · ×3



1/4-20 × 3-1/2" black-oxide connector bolts (Rockler 49153) · 1/4" x 3-1/2" · ×4



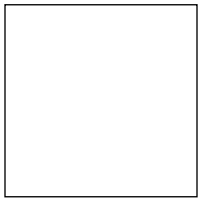
#10 × 2" pan-head wood screws and washers · 3/16" x 2" · ×8



#8 × 1-1/2" flat-head wood screws · 5/32" x 1-1/2" · ×52



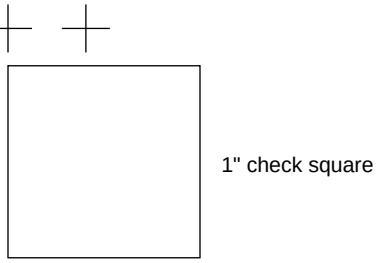
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1" check square

Fasteners, Actual Size · page A2 (1 row x 2)

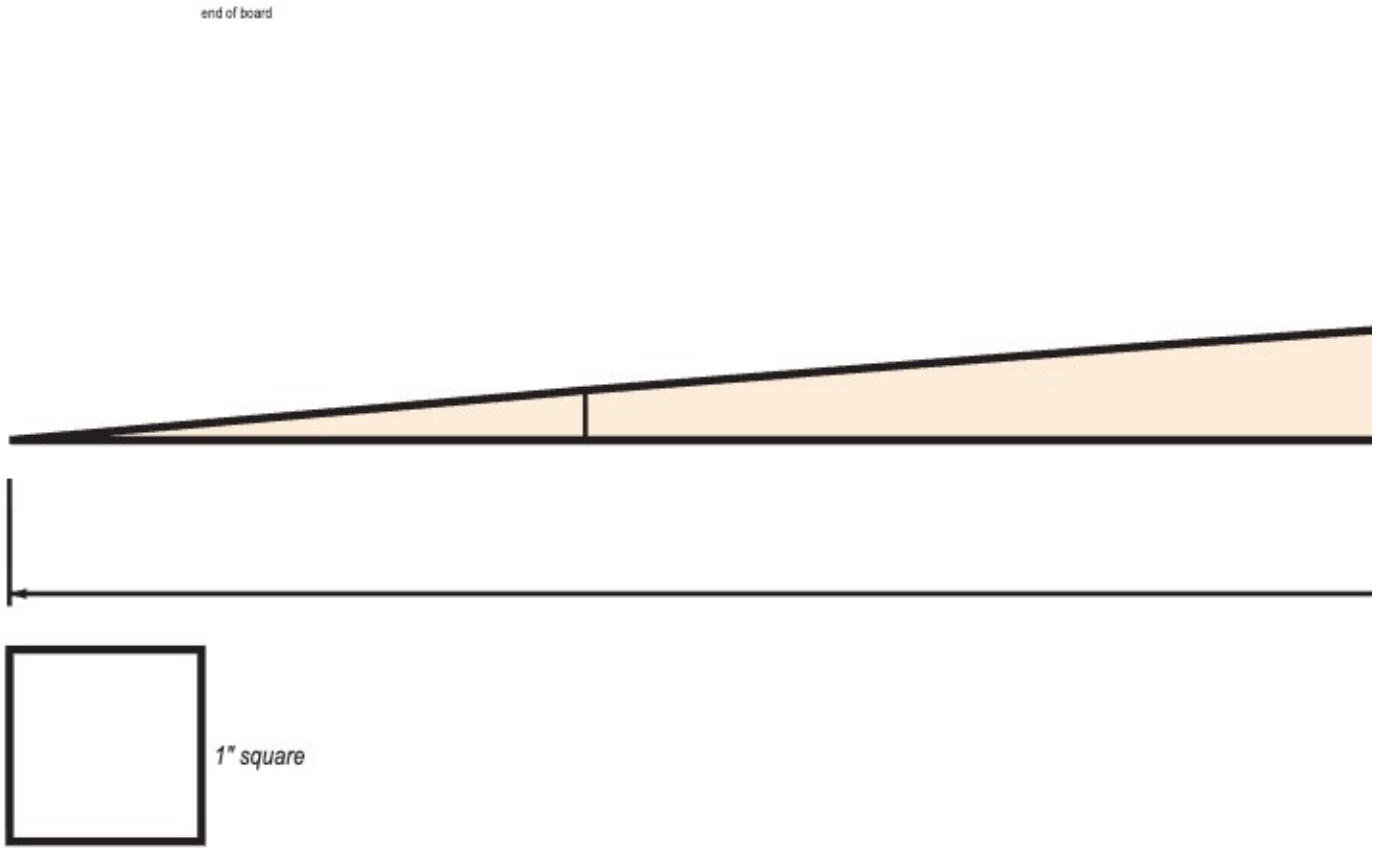
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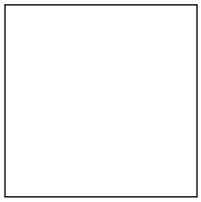


1" check square

Headboard Curve Template · page A1 (1 row x 5)

FULL SIZE: print at 100% ("Actual size"). Overlap 1/4" with the next page and match the marks.

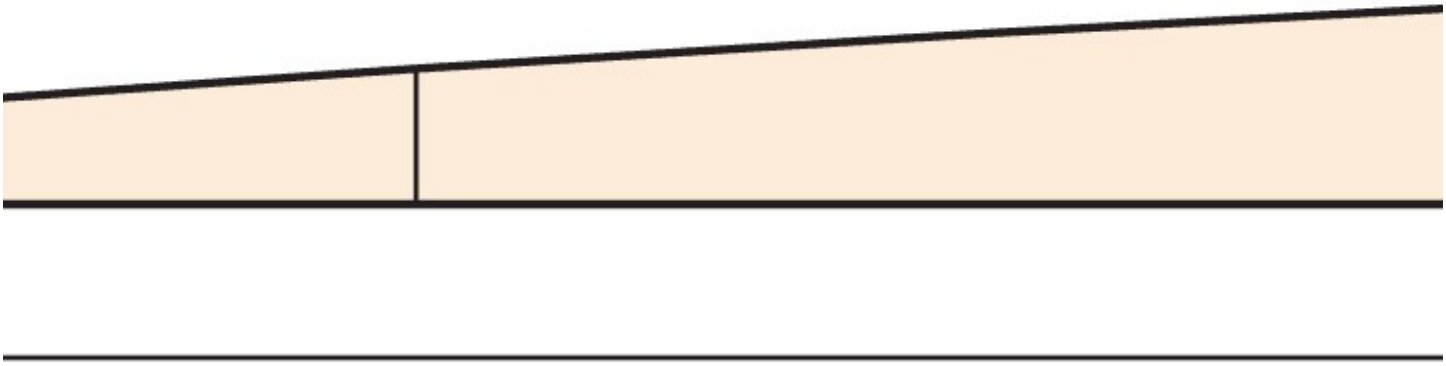




1" check square

Headboard Curve Template · page A2 (1 row x 5)

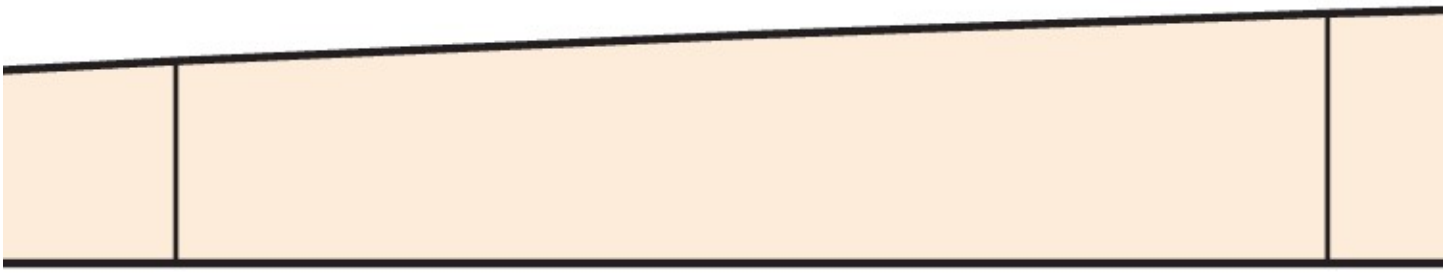
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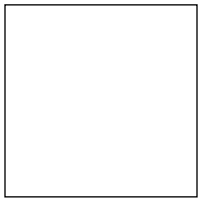
1" check square

Headboard Curve Template · page A3 (1 row x 5)

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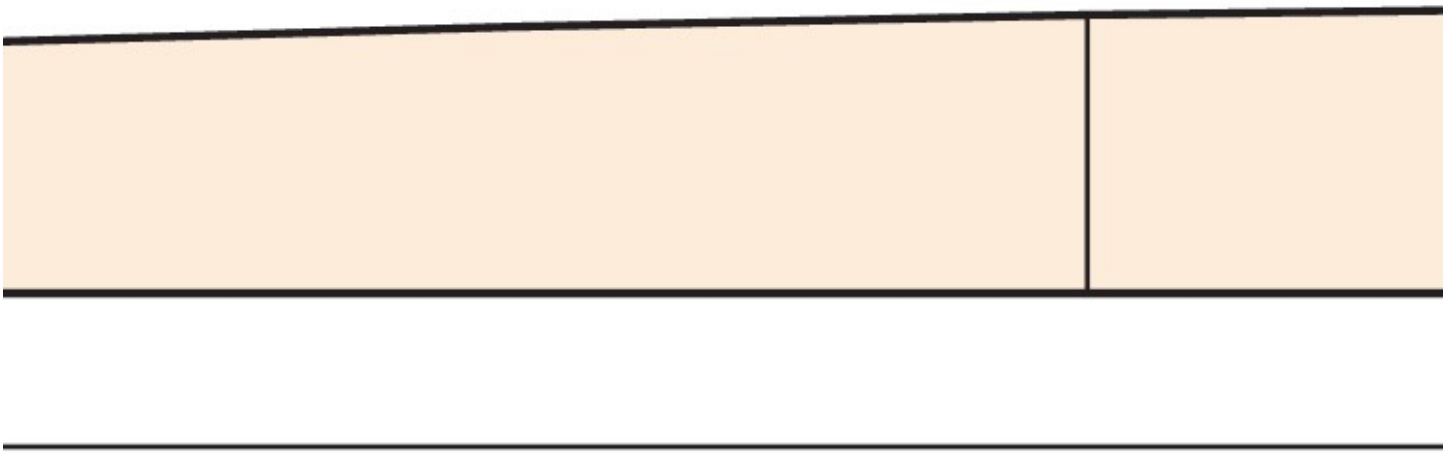
33"



1" check square

Headboard Curve Template · page A4 (1 row x 5)

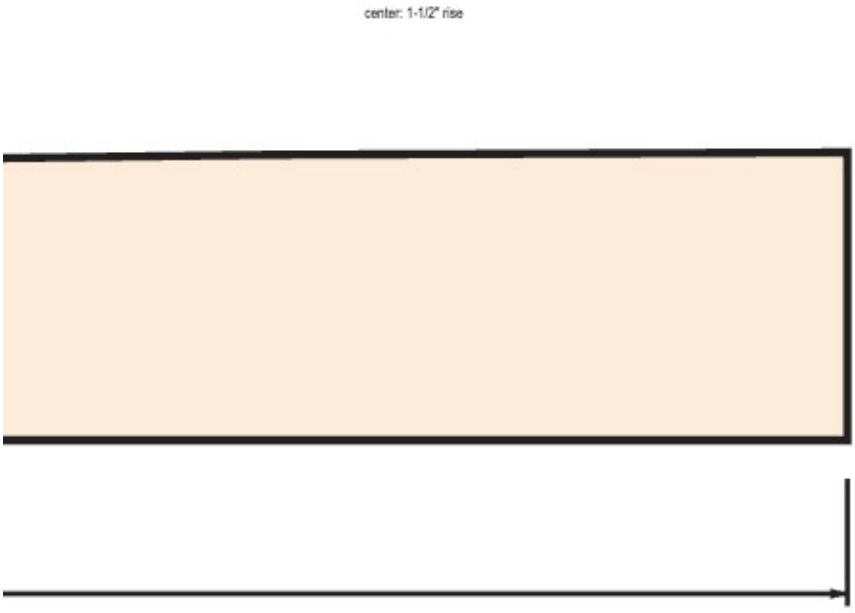
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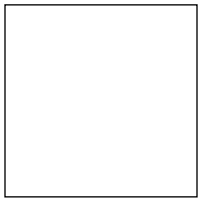


1" check square

Headboard Curve Template · page A5 (1 row x 5)

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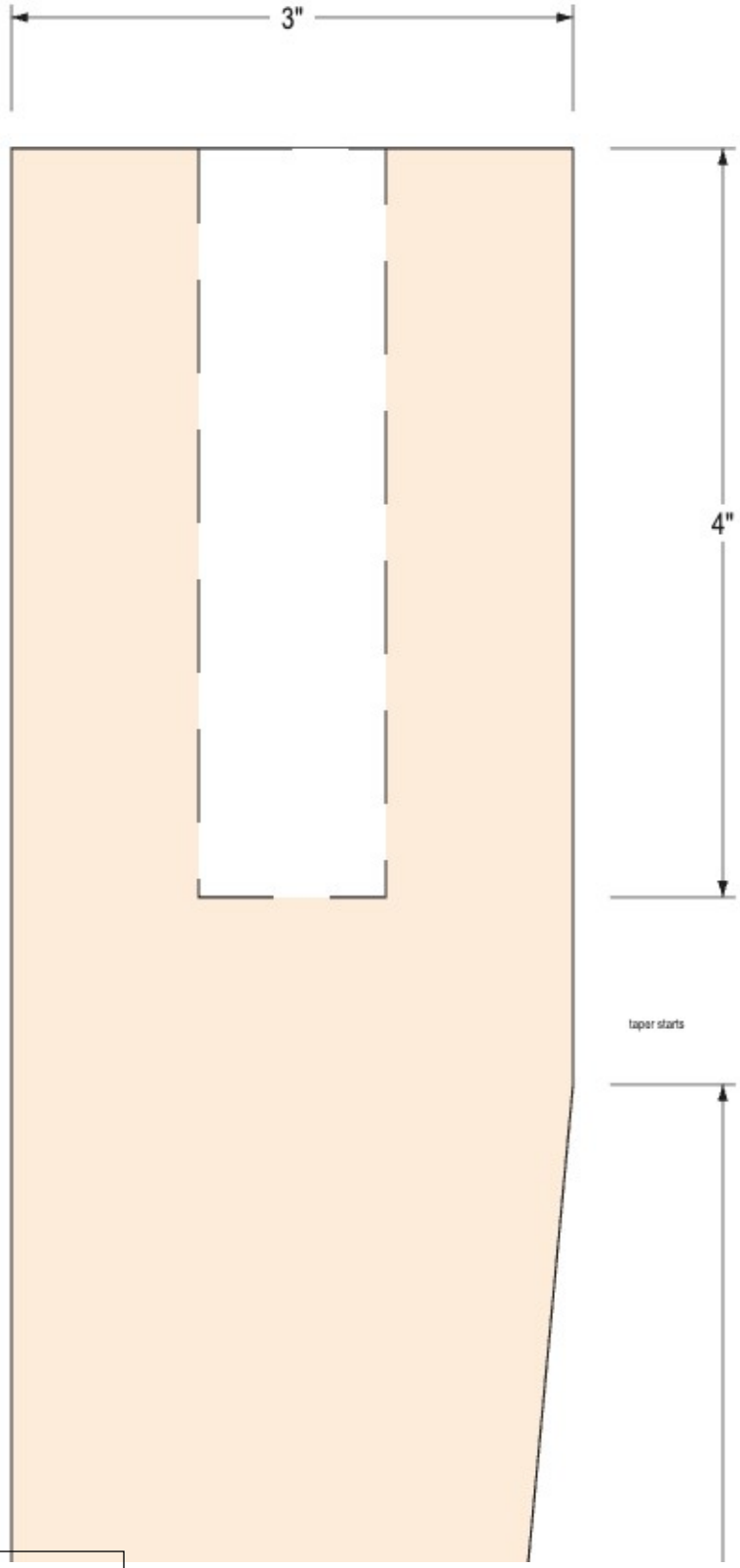


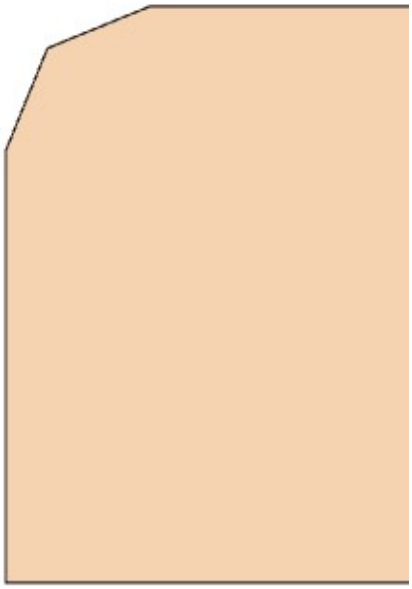


1" check square

Leg Pattern · page A1 (2 rows x 2)

FULL SIZE: print at 100% ("Actual size"). Overlap 1/4" with the next page and match the marks.



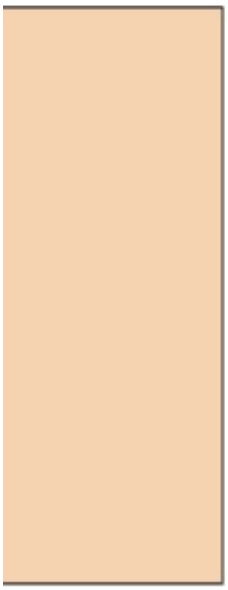


top: 3/4" R outside corner

1" check square

Leg Pattern · page A2 (2 rows x 2)

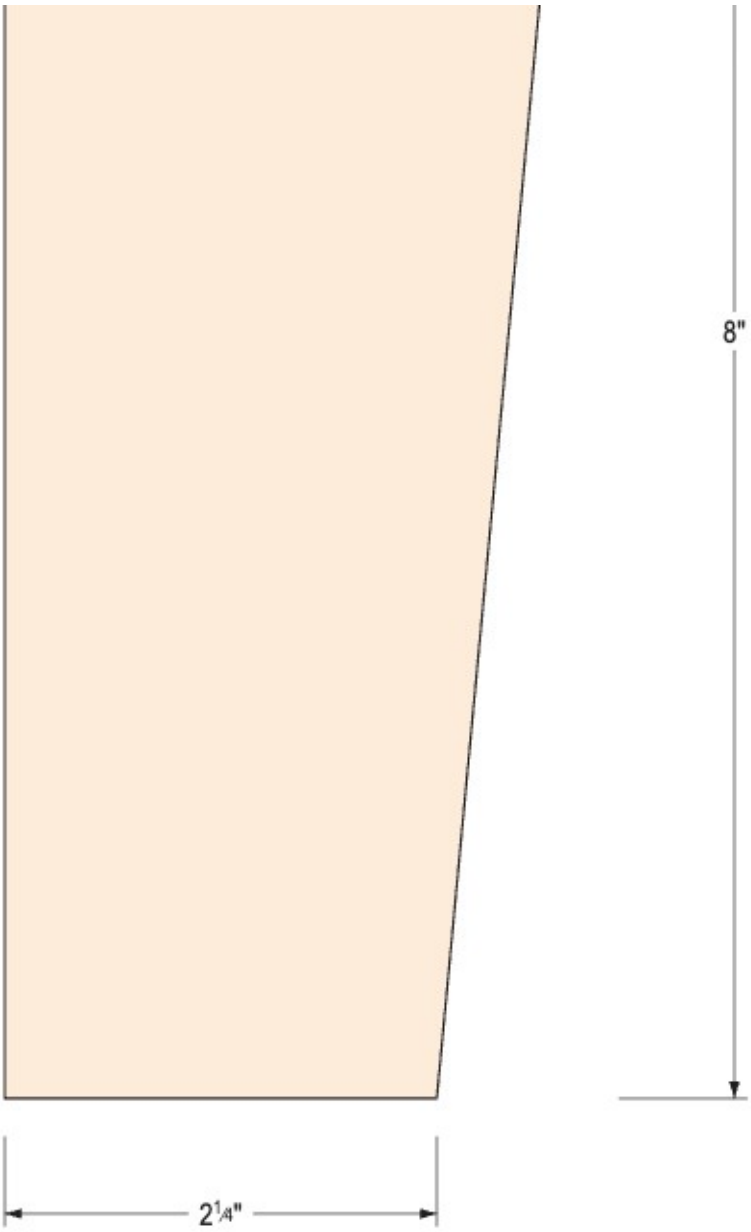
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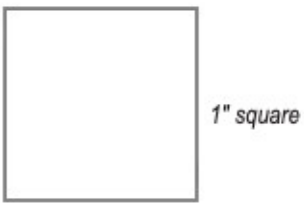


1" check square

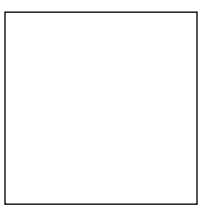
Leg Pattern · page B1 (2 rows x 2)

FULL SIZE: print at 100% ("Actual size"). Overlap 1/4" with the next page and match the marks.





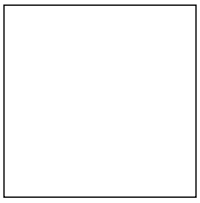
1" square



1" check square

Leg Pattern · page B2 (2 rows x 2)

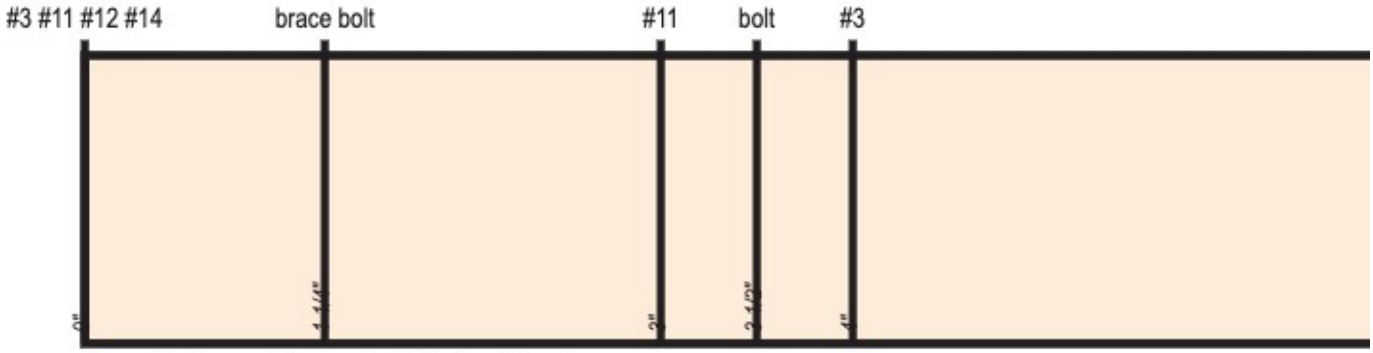
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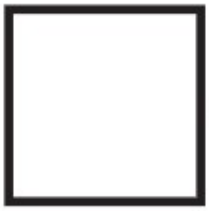
1" check square

Headboard Story Stick · page A1 (1 row x 5)

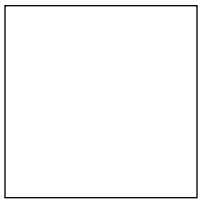
FULL SIZE: print at 100% ("Actual size"). Overlap 1/4" with the next page and match the marks.



From the floor, at the ends of the headboard.



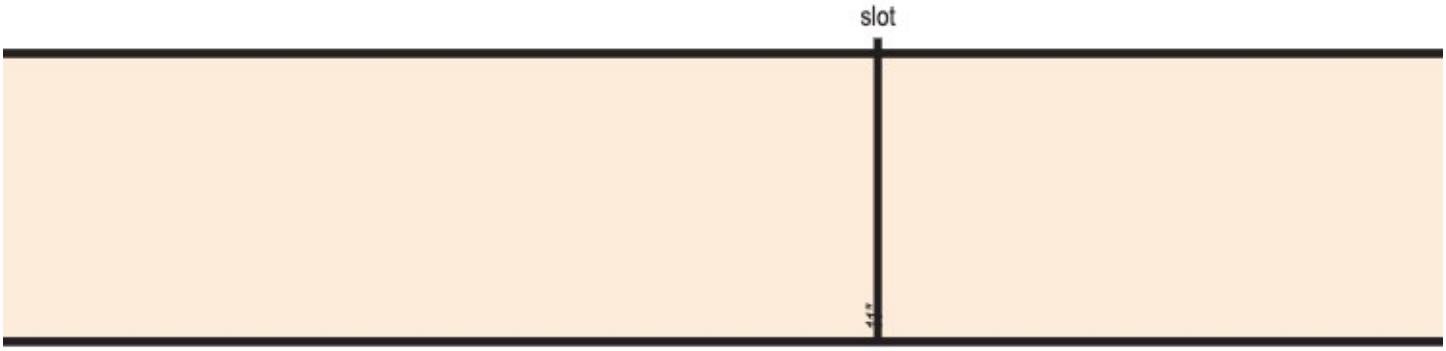
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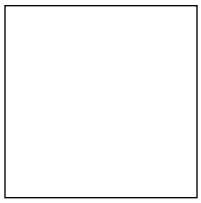


1" check square

Headboard Story Stick · page A2 (1 row x 5)

FULL SIZE: print at 100% ("Actual size"). Overlap 1/4" with the next page and match the marks.





1" check square

Headboard Story Stick · page A3 (1 row x 5)

FULL SIZE: print at 100% ("Actual size"). Overlap 1/4" with the next page and match the marks.



1" check square

Headboard Story Stick · page A4 (1 row x 5)

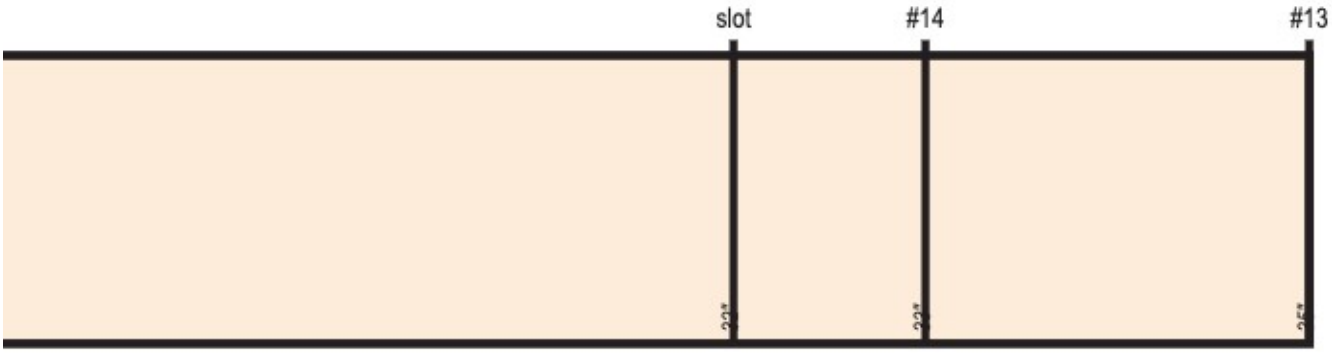
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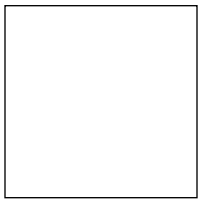


1" check square

Headboard Story Stick · page A5 (1 row x 5)

FULL SIZE: print at 100% ("Actual size"). Overlap 1/4" with the next page and match the marks.





1" check square